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1 Introduction

The lithium-ion battery is the predominant form of commercial rechargeable batteries, and are wildy used in many applications

Lithium-ion batteries are use in many wide spread applications, such as consumer electronics, toys, power tools, and electric vehicles

2 Background

The capacity of a lithium-ion battery — the total charge deliverable in a full discharge — fades with repeated cycling. With time and use, storage capacity diminishes and internal resistance increases, driven by degradation mechanisms that can occur simultaneously or trigger one another [2]. A battery is said to reach end-of-life (EoL) when its capacity drops below 80% of its initial value, following the criterion used by Zhou et al. [7].

The rate at which a battery charges or discharges is characterized by the C-rate, defined as the ratio of applied current to nominal capacity [4]. At 1C, a full charge or discharge completes in one hour; at 2C, in thirty minutes. Higher C-rates induce greater mechanical stress during cycling and promote lithium plating during charge [2], both of which accelerate capacity fade. In the dataset used in this study, all cells are discharged at a fixed 4C rate while the charging protocol varies — making the average charging C-rate the primary source of variability in operating conditions across cells.

Data Overview

The primary data for this study is sourced from the battery cycling datasets hosted on `data.matr.io` [6].. The consolidated dataset consists of 169 Lithium-ion phosphate (LFP) cells cycled under varying fast-charging conditions. These cells represent 78 distinct charging protocols and have lifetimes that vary accordingly.

Data Structure and Extraction

The raw data, originally stored in MATLAB format, was ingested into a Python-based workflow following the methodology from Attia et al.'s GitHub [1]. The dataset is organized hierarchically: at the cell level, each battery is indexed by a unique key and carries a charging protocol string and a cycle life — the number of cycles until end-of-life (EoL) is reached. Below this, cycle-by-cycle summary measurements (capacity, internal resistance, temperature) and intra-cycle time-series (voltage, current, capacity curves) are accessible for each cell. A full description of available variables is provided in the Appendix.

Data Partitioning and Grouping Logic

Zhou and Howey (2023) partitioned the data into eight groups based on their average C-rate and calculated the variance partition coefficient (VPC). Specifically, they calculated:

$$\mathrm{VPC} = \frac{\sigma^2_{\text{group}}}{\sigma^2_{\text{individual}} + \sigma^2_{\text{group}}}$$

$$\sigma^2_{\text{group}} = \mathrm{var}\left(\bar{u}_j - \bar{u}\right), \quad \bar{u} = \frac{1}{N}\sum_{j=1}^{k}\sum_{i=1}^{n_j} y_{ji}$$

$$\sigma^2_{\text{individual}} = \mathrm{var}\left(y_{ji} - \bar{u}_j\right), \quad \bar{u}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} y_{ji}$$

where σ^2_{group} represents lifetime variance between groups and $\sigma^2_{\text{individual}}$ represents the total within-group lifetime variance. They obtained a VPC of 74.5%, implying that between-group variance accounts for the majority of total lifetime variance. It follows that any statistical model which ignores group-level variation would be discarding useful information about the data's variability, while fitting separate statistical models on each group would ignore the remaining within group variability. A hierarchical Bayesian model addresses both by partially pooling estimates across groups.

Grouping Results

Following Zhou and Howey [7], we first applied constrained K-means clustering on average charging C-rate. While this produced groups with low within-group C-rate variance (see Figure 1), the resulting VPC was only 38.973%. We subsequently grouped cells by their unique charging policy string and optimized the grouping via a hill-climbing algorithm — iteratively reassigning policy strings across the eight groups to maximise VPC — achieving 74.406The hill-climb grouping exhibits higher within-group C-rate variance but lower within-group EoL variance (see Figure 2).

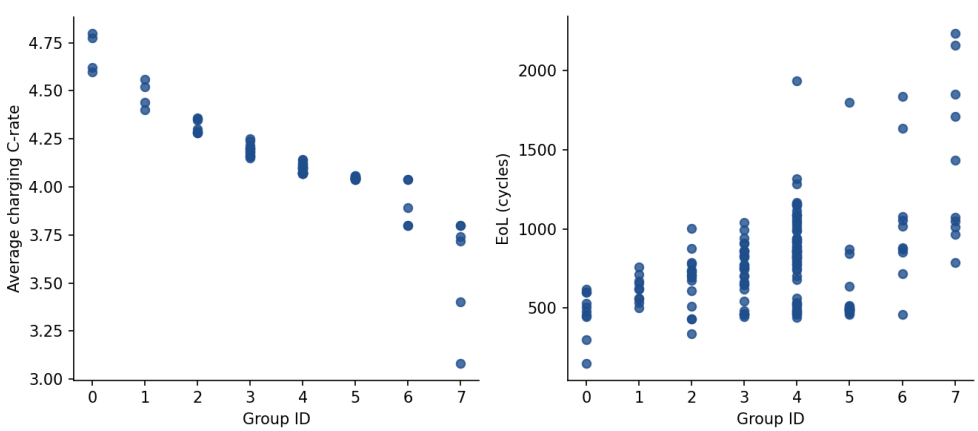

Figure 1: Distribution of average charging C-rate and EoL for each group obtained from constrained K-means.

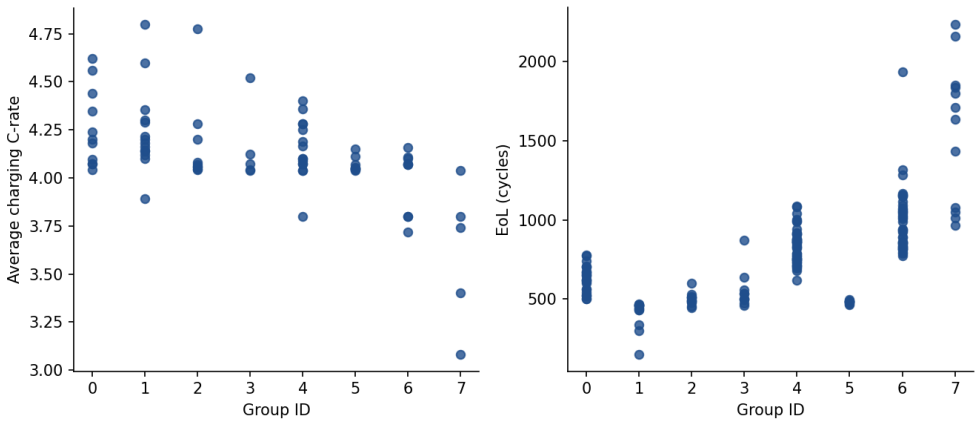

Figure 2: Distribution of average charging C-rate and EoL for each group obtained from hill-climb optimization.

Introduction to Hierarchical Bayes

Multilevel Modelling

To motivate hierarchical modelling, consider first the standard linear regression setting:

$$y = \beta_0 + \beta_1 x_i + \epsilon$$

Now suppose the data naturally contains J groups — for example, homes prices across J states, where the relationship between price and square footage may differ by state. This suggests allowing group-specific parameters:

$$y_i = \alpha_{j[i]} + \beta_{j[i]} x_i + \epsilon_i$$

where j is the group index and $(\alpha_{j[i]}, \beta_{j[i]})$ follow some distribution. For example, they may be jointly normally distributed [3]:

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \mathbf{\Sigma} \right), \text{ for } j = 1, \ldots, J$$

Pooling Tradeoff

In the example above, the multilevel regression may use the same parameters for all groups (complete pooling), or fit entirely separate models for each group with no information shared across them (no pooling). In the latter case, the parameter distribution is specified independently for each group:

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_{\alpha,j} \\ \mu_{\beta,j} \end{pmatrix}, \mathbf{\Sigma}_j \right), \quad j = 1, \ldots, J$$

Pooling Tradeoff: County Radon Levels

Gelman et al. [3] analyzed a dataset consisting of radon levels for the 85 counties of Minnesota. The objective was to estimate the distribution of the radon levels of the houses within each county. The first proposed approach was to calculate statistics (e.g., average) using all homes from the sample

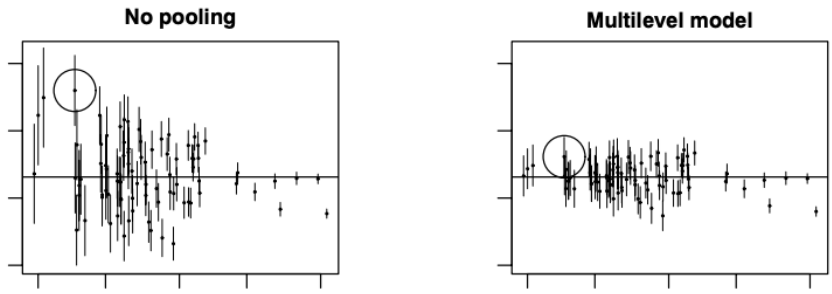

Figure 3: Estimates ± standard errors for average log radon levels in Minnesota counties versus number of observations per county: (a) no-pooling, (b) partial pooling. Counties with fewer observations have more variable estimates and larger standard errors under no pooling, illustrating how small samples produce unreliable county-level estimates.

(complete pooling). This approach does not consider variability among counties and complicates the task of estimating county-specific distributions.

The second proposed approach was to calculate the average log-radon levels in each county (no pooling). With this approach, the authors found that the highest county-specific average log radon level of 85 was much larger than the values from different counties. However, this outlier may not be representative of the actual average log radon levels of its county, since it was obtained from a sample of only two observations. No pooling therefore produces unreliable estimates for data-sparse groups, motivating a middle ground.

Partial Pooling

Gelman et al. [3] provided a middle ground between complete and no pooling, which they called partial pooling. Specifically, they defined the observation-level model as:

$$y_i \sim N(\alpha_{j[i]} + \beta x_i, \sigma_y^2), \quad i = 1, \ldots, n$$

with the group-level model:

$$\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad j = 1, \ldots, J$$

Together they form a two-level structure: in the first equation, each observation is normally distributed around its group's intercept plus a shared slope; in the second, the group intercepts are drawn from a common population distribution.

The two-level structure creates dependency between group-level intercepts through the shared prior $N(\mu_\alpha, \sigma_\alpha^2)$. Specifically, each α_j is estimated from its group's data and also influenced by all other groups. This pulls data-sparse groups toward the population mean μ_α while leaving data-rich groups largely unaffected. The strength of the pull is also determined by σ_α, which is learned from the data.

Hierarchical Bayes

In the multilevel modelling example above, the parameters μ_α and σ_α^2 are unknown. In Bayesian statistics, these parameters are treated as random variables which are assigned prior distributions. Placing priors on these parameters gives the full hierarchical Bayesian model:

Level 3 (Hyperpriors): $\mu_\alpha \sim p(\mu_\alpha), \quad \sigma_\alpha^2 \sim p(\sigma_\alpha^2)$

Level 2 (Group prior): $\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad j = 1, \ldots, J$

Level 1 (Observation model): $y_i \sim N(\alpha_{j[i]} + \beta x_i, \sigma_y^2), \quad i = 1, \ldots, n$

Hierarchical Bayesian Model for Battery Lifetime Prediction

Model Details

Following Zhou & Howey [7], we instantiate the three-level HBM for battery lifetime prediction. The response variable y_{ji} denotes the EoL (in days) for cell i in group j. At the individual level, each cell is represented by a feature vector $\mathbf{x}_{ji} \in \mathbb{R}^4$ constructed from early-cycle measurements; at the group level, each cycling condition group is represented by $\mathbf{g}_j \in \mathbb{R}^2$, which encodes the group's average C-rate. A full description of model variables and features is provided in the Appendix.

Level 3 (Hyperpriors): $\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, 10\mathbf{I})$

Level 2 (Cycling conditions): $\boldsymbol{\theta}_j \sim \mathcal{N}(\boldsymbol{\gamma}^\top \mathbf{g}_j, \sigma_j^2), \quad j = 1, \ldots, J$

Level 1 (Individual cells): $y_{ji} \sim \mathcal{N}(\boldsymbol{\theta}_j^\top \mathbf{x}_{ji}, \sigma_y^2), \quad i = 1, \ldots, n_j$

The three-level structure encodes two sources of variability simultaneously: individual cell variability within each group (Level 1) and variability in how features relate to EoL across cycling conditions (Level 2). Crucially, at Level 2, the group coefficients $\boldsymbol{\theta}_j$ are not estimated independently — they are drawn from a common distribution centred at $\boldsymbol{\gamma}^\top \mathbf{g}_j$, allowing for information sharing across groups.

Inference

Inference follows a sequential process. First, the posterior over the hyperparameters is estimated via MCMC [7]:

$$P(\boldsymbol{\gamma} \mid \{Y\}) \propto P(\boldsymbol{\gamma}) \prod_{j=1}^{J} P(Y_j \mid \boldsymbol{\gamma})$$

where $P(\boldsymbol{\gamma}) = \mathcal{N}(\mathbf{0}, 10\mathbf{I})$ is the hyperprior. Since the group parameters $\boldsymbol{\theta}_j$ are unknown at this stage, they are treated as random variables and marginalised out of the likelihood:

$$P(Y_j \mid \boldsymbol{\gamma}) = \int P(Y_j \mid \boldsymbol{\theta}_j) \cdot P(\boldsymbol{\theta}_j \mid \boldsymbol{\gamma})\, d\boldsymbol{\theta}_j$$

The resulting posterior $P(\boldsymbol{\gamma} \mid \{Y\})$ has no closed form due to the intractable normalising constant $P(\{Y\})$; MCMC is therefore used to obtain samples from it. These samples are then used as a prior to obtain the group-level posteriors $P(\boldsymbol{\theta}_j \mid \boldsymbol{\gamma}, Y_j)$ via Bayes' rule [7]. Starting from the joint distribution, we have:

$$P(\boldsymbol{\theta}_j, \gamma, Y_j) = P(Y_j \mid \boldsymbol{\theta}_j) \cdot P(\boldsymbol{\theta}_j \mid \gamma) \cdot P(\gamma)$$

Conditioning on γ and Y_j and normalizing gives:

$$P(\boldsymbol{\theta}_j \mid \gamma, Y_j) = \frac{P(Y_j,\, \boldsymbol{\theta}_j,\, \gamma)}{P(Y_j \mid \gamma)} \tag{1}$$

$$= \frac{P(Y_j \mid \boldsymbol{\theta}_j,\, \gamma) \cdot P(\boldsymbol{\theta}_j,\, \gamma)}{P(Y_j \mid \gamma)} \tag{2}$$

$$= \frac{P(Y_j \mid \boldsymbol{\theta}_j) \cdot P(\boldsymbol{\theta}_j \mid \gamma) \cdot P(\gamma)}{P(Y_j \mid \gamma)} \tag{3}$$

where in the third line we use the fact that once $\boldsymbol{\theta}_j$ is known, γ carries no additional information about Y_j, so $P(Y_j \mid \boldsymbol{\theta}_j,\, \gamma) = P(Y_j \mid \boldsymbol{\theta}_j)$.

2.1 Bayesian Heirarchical Models

A hierarchical model is a statistical model that includes multiple levels of parameters, where the parameters at one level are treated as random variables that depend on parameters at a higher level. In a Bayesian hierarchical model, we specify prior distributions for the parameters at each level, and use Bayes' theorem to update our beliefs about these parameters based on observed data.

We would like to capture both individual battery characteristics and the influence of different usage (cycling) conditions. A two level hierarchical structure is considered:

- Level 1 (Cell-level parameters θ_j): Each battery j has its own parameters θ_j that represent how early cycle features relate to battery aging for that particular subgroup or cycling condition.
- Level 2 (Hyperparameters γ): These represent the parameters that govern the distribution of cell-level parameters across all subgroups, meaning they capture similarities across different batteries and cycling conditions.

Bayes rule is used to combine prior knowledge with the observed early-cycle data from multiple batteries to estimate posterior distributions of parameters at both levels.

The level 2 posterior is calculated as

$$P(\gamma|\{Y\}) \propto \text{Likelihood} \times \text{Prior}$$

2.2 Multichain Monte Carlo

References

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Appendix

Data Dictionary

Table 1: Cell Level Variables

Variable	Description	Type
`cycle_life`	Number of cycles until EoL	Float
`charge_policy`	Charging protocol	String
`summary`	Metadata for each cycle	Dictionary
`cycles`	Time series of measurements	Dictionary

Table 2: Summary Level (Cycle-by-Cycle Metadata)

Variable	Description	Type
`IR`	Internal resistance	Numeric array
`QC`	Charge capacity	Numeric array
`QD`	Discharge capacity	Numeric array
`Tavg`	Average temperature	Numeric array
`Tmin`	Minimum temperature	Numeric array
`Tmax`	Maximum temperature	Numeric array
`chargetime`	Charge time	Numeric array
`cycle`	Cycle number	Numeric array

Table 3: Cycle Level (Intra-cycle Time Series)

Variable	Description	Type
`I`	Current	Numeric array
`Qc`	Charge capacity	Numeric array
`Qd`	Discharge capacity	Numeric array
`Qdlin`	Linear interpolated discharge capacity	Numeric array
`T`	Temperature	Numeric array
`Tdlin`	Linear interpolated temperature	Numeric array
`V`	Voltage	Numeric array
`dQdV`	Derived vectors of discharge	Numeric array
`t`	Time	Numeric array

Model Variables and Features

Table 4: Variables

Symbol	Description
$j = 1, \ldots, J$	Cycling condition group index
$i = 1, \ldots, n_j$	Cell index within group j
y_{ji}	EoL time (days) for cell i in group j
$\boldsymbol{\theta}_j \in \mathbb{R}^4$	Regression coefficients for group j
σ_j^2	Variance of group j model
$\boldsymbol{\gamma} \in \mathbb{R}^{4\times 2}$	Hyperparameter matrix linking cycling conditions to group coefficients

Table 5: Feature vectors

Symbol	Components	Level
$\mathbf{x}_{ji} \in \mathbb{R}^4$	$(1, F1_{ji}, F2_{ji}, F3_{ji})^\top$	Individual
$\mathbf{g}_j \in \mathbb{R}^2$	$(1, \bar{g}_j)^\top$	Group

Table 6: Individual-level features (computed from first 100 cycles)

Symbol	Description
$F1_{ji}$	Variance of $\Delta Q(V)$ between cycles 10 and 100
$F2_{ji}$	Minimum of $\Delta Q(V)$ over the same interval
$F3_{ji}$	Discharge capacity at cycle 2

[1]

[1] $\Delta Q(V)$ denotes the discharge curve difference as a function of voltage, following Severson et al. (2019) [5].